

Name: Solutions

Teacher:



**Sydney
Secondary
College**
Leichhardt

Year 9 Stage 5.1/5.2/5.3

Mathematics

Yearly Exam

2022

Instructions

- Working time: **50 minutes**
- Attempt all questions
- Write in black or blue pen, draw diagrams, tables and graphs in pencil
- Show relevant working out where necessary

	Surface Area And Volume	Trigonometry	Probability	Total
5.2	/15	/15	/15	/45
5.3	/15	/15	/15	/45

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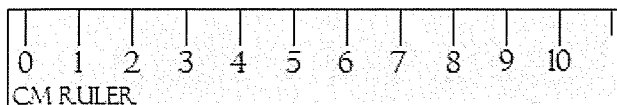
Multiple Choice (1 mark each) – CLEARLY circle the correct letter, A, B, C or D.

Surface Area And Volume

1. _____ is the distance around a two-dimensional shape.

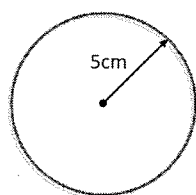
- a) ☒ Perimeter b) Area c) Volume d) Capacity

2. What are the limits of accuracy of this ruler?



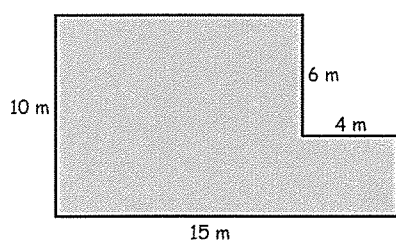
- a) $\pm 0.5 \text{ mm}$ b) $\pm 1 \text{ mm}$ c) ☒ $\pm 0.5 \text{ cm}$ d) $\pm 1 \text{ cm}$

3. Which of the following expression gives the area of the circle shown?



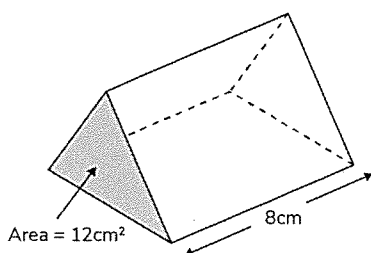
- a) $2 \times \pi \times 5$ b) $2 \times \pi \times 5^2$ c) ☒ $\pi \times 5^2$ d) $\pi \times 10^2$

4. The perimeter of the shape shown below is:



- a) 35 m b) 40 m c) 45 m d) ☒ 50 m

5. What is the volume of the triangular prism shown?



- a) 72 cm^3 b) 84 cm^3 c) ☒ 96 cm^3 d) 108 cm^3

Section 1 continues on the following page

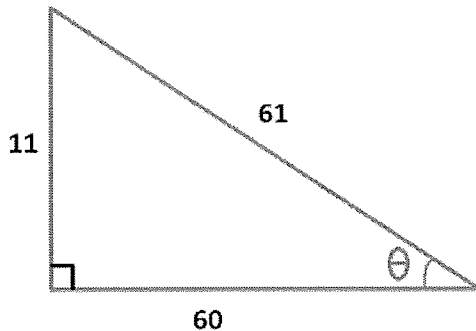
Multiple Choice (1 mark each) – CLEARLY circle the correct letter, A, B, C or D.

Trigonometry

6. The angle $46^{\circ}32'53''$ rounded to the nearest degree is:

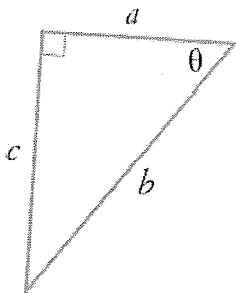
- a) $46^{\circ}32'$ b) 46° ☒ c) 47° d) $46^{\circ}33'$

7. For the triangle below, using θ as the reference angle, the length of the adjacent side is:



- ☒ a) 60 b) 61 c) 11 d) $\frac{11}{60}$

8. In the triangle below, $\sin \theta =$



- a) $\frac{c}{a}$ b) $\frac{a}{b}$ c) $\frac{a}{c}$ ☒ d) $\frac{c}{b}$

9. $15 \sin 54^{\circ}$ correct to two decimal places is:

- ☒ a) 12.14 b) 0.81 c) 10.61 d) 0.71

10. If $\cos \theta = 0.5$, what is the value of θ ?

- a) 30° b) 45° ☒ c) 60° d) 90°

Section 1 continues on the following page

Multiple Choice (1 mark each) – CLEARLY circle the correct letter, A, B, C or D.

Probability

11. The _____ is the set of all possible outcomes in a chance experiment.

- a) Event b) Trial ☒ c) Sample Space d) Complement

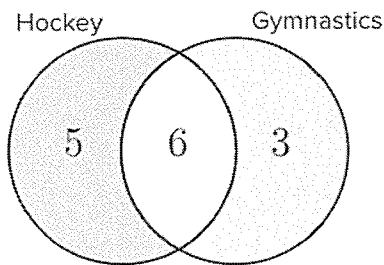
12. A letter is selected at random from the word TRIGONOMETRY. The probability of selecting “the letter T” is:

- a) $\frac{1}{3}$ b) $\frac{1}{4}$ c) $\frac{1}{5}$ ☒ d) $\frac{1}{6}$

13. A six-sided die is rolled 120 times. What is the expected number of times rolling a “5 or 6”?

- a) 20 ☒ b) 40 c) 60 d) 80

14. The Venn diagram shows the number of students who play either hockey or gymnastics. How many students play hockey and gymnastics?



- a) 5 ☒ b) 6 c) 3 d) 8

15. The two-way table shows the types of computers owned by 100 people. How many people do not have a Mac?

	Mac	No Mac	Total
PC	12	50	62
No PC	31	7	38
Total	43	57	100

- a) 43 ☒ b) 57 c) 62 d) 38

Short Answer Questions – Show working where necessary.

Surface Area And Volume

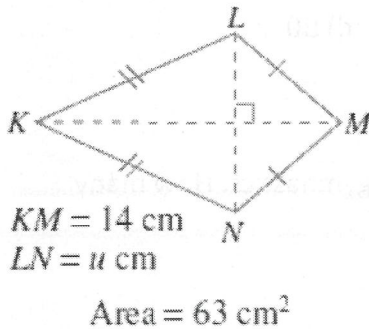
1. If the average file size of a photo is 2.5 MB, find the number of photos that can be stored on a 1 GB USB flash drive. 1

$$1 \text{ GB} = 1000 \text{ MB}$$

$$\frac{1000}{2.5} = 400$$

400 photos ①

2. Find the length of the diagonal LN if the kite has an area of 63 cm^2 and the other diagonal KM has a length of 14 cm. 1

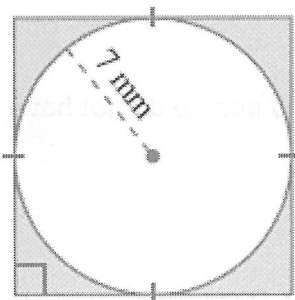


$$\frac{1}{2} \times u \times 14 = 63$$

$$u = \frac{63}{\frac{1}{2} \times 14}$$

$$= 9 \text{ cm } ①$$

3. Find the area of the shaded part, correct to one decimal place. 2



$$\begin{aligned} \text{Area of square} &= 14 \times 14 \\ &= 196 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area of circle} &= \pi \times 7^2 \\ &= 153.93804 \end{aligned}$$

$$\begin{aligned} \text{Shaded part} &= 196 - 153.93804 \\ &= 42.1 \text{ mm}^2 \end{aligned}$$

① Finding either area of square or circle

① Correct answer

or

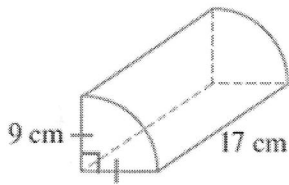
② Correct answer

Section 2 continues on the following page

Short Answer Questions – Show working where necessary.

4. Find the total surface area of the solid, correct to one decimal place.

2



$$\begin{aligned}
 & \frac{2\pi r^2 + 2\pi rh}{4} + (9 \times 17 \times 2) \\
 &= \frac{2 \times \pi \times 9^2 + 2 \times \pi \times 9 \times 17}{4} + 306 \\
 &= 367.5663405 + 306 \\
 &= 673.6 \text{ cm}^2
 \end{aligned}$$

① Working out
 ① Correct answer
or
 ② Correct answer

5. Find the side length of a cube whose surface area is 3456 cm^2 .

2

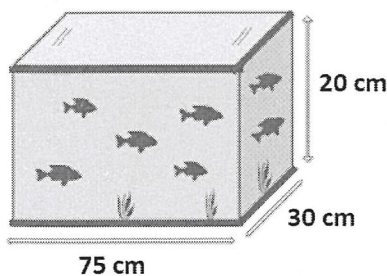
Let side length = x

$$\begin{aligned}
 x \times x \times 6 &= 3456 \\
 6x^2 &= 3456 \\
 x^2 &= 576 \\
 x &= 24 \text{ cm}
 \end{aligned}$$

① Working out
 ① Correct answer
or
 ② Correct answer

6. 28 L of water is poured into a rectangular fish tank of dimensions 75 cm by 30 cm by 20 cm. How many more litres (L) of water is required to fill the fish tank to the top?

2



$$\begin{aligned}
 75 \times 30 \times 20 &= 45000 \text{ cm}^3 \\
 &= 45000 \text{ mL} \\
 &= 45 \text{ L}
 \end{aligned}$$

$$45 - 28 = 17 \text{ L}$$

17 more litres is required

① Working out
 ① Correct answer
or
 ② Correct answer

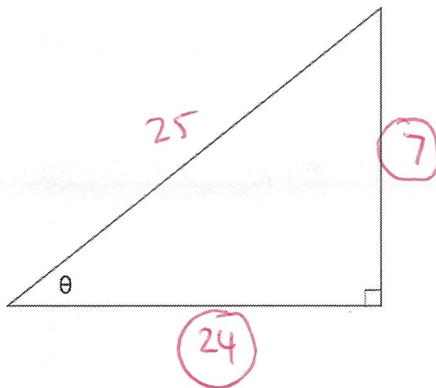
Short Answer Questions – Show working where necessary.

Trigonometry

7. A right-angled triangle has the trigonometric ratio $\tan \theta = \frac{7}{24}$.

a) Show this information on the triangle.

1



① For 7 and 24
on correct sides

b) Find the ratio for $\sin \theta$.

2

$$\begin{aligned} \text{Hypotenuse} &= \sqrt{7^2 + 24^2} \\ &= 25 \end{aligned}$$

$$\sin \theta = \frac{7}{25}$$

① working out

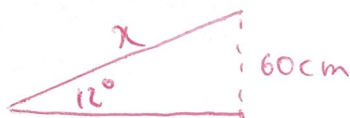
① Correct answer

or

② Correct answer

8. The entrance to a shopping centre is 60 cm above the ground level. A wheelchair ramp is built to the entrance at an angle of 12° with the ground. Calculate (in cm) the length of the ramp, correct to one decimal place.

2



$$\sin 12 = \frac{60}{x}$$

$$x = \frac{60}{\sin 12}$$

$$= 288.6 \text{ cm}$$

① working out

① Correct answer

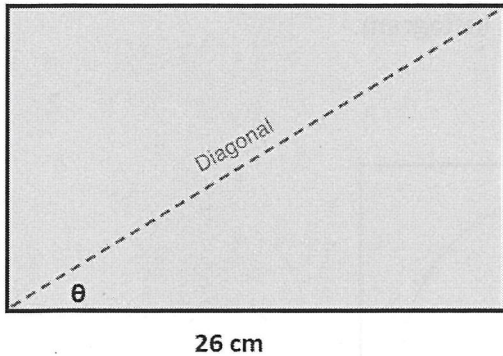
or

② Correct answer

Section 2 continues on the following page

Short Answer Questions – Show working where necessary.

9. A rectangle has an area of 312 cm^2 and a length of 26 cm . Find the angle θ , formed between the length and the diagonal, correct to the nearest degree. 2



$$26 \times \text{Width} = 312$$

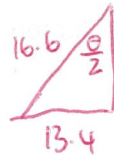
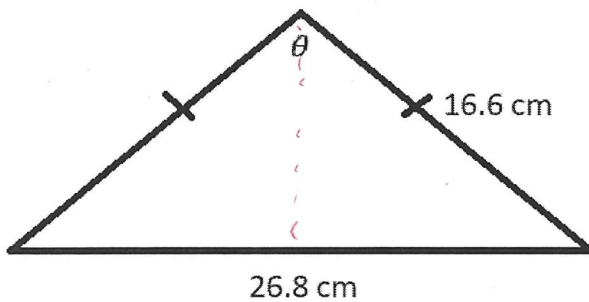
$$\text{Width} = 12 \text{ cm}$$

$$\tan \theta = \frac{12}{26}$$

$$\theta = \tan^{-1} \left(\frac{12}{26} \right)$$

$$= 25^\circ$$

10. Find the apex angle, θ , of an isosceles triangle if the length of the equal sides is 16.6 cm and the length of the base is 24.8 cm . Give the answer correct to the nearest minute. 2



$$\sin \frac{\theta}{2} = \frac{13.4}{16.6}$$

$$\frac{\theta}{2} = \sin^{-1} \left(\frac{13.4}{16.6} \right)$$

$$= 53.82606584$$

$$\theta = 53.82606584 \times 2$$

$$= 107^\circ 39'$$

- ① working out
① correct answer

on

- ② correct answer

11. If $\sin \theta = 0.5432$, find $\cos \theta$, correct to two decimal places. 1

$$0.84$$

Section 2 continues on the following page

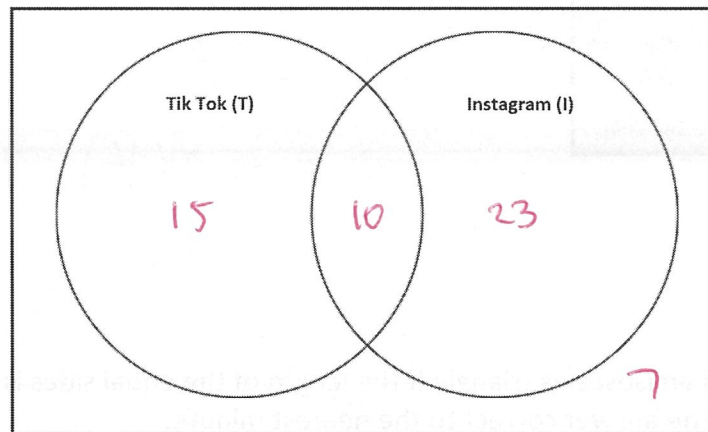
Short Answer Questions – Show working where necessary.

Probability

12. A survey of 55 students found that 25 students like Tik Tok (T) and 33 students like Instagram (I). 10 students like both Tik Tok and Instagram and 7 like neither Tik Tok nor Instagram.

a) Fill in the Venn diagram to represent the survey results.

2



① 2 or 3 values correct

② All values correct

b) If one of the 55 students was randomly selected, find:

i. $P(\text{likes neither Tik Tok nor Instagram})$

1

$$\frac{7}{55}$$

ii. $P(\text{likes Tik Tok})$

1

$$\frac{25}{55} = \frac{5}{11}$$

13. The two-way table shows data collected about Year 9 Students.

	Left-handed	Right-handed
Male	12	55
Female	10	43

a) What is the percentage probability that a student selected at random is male and left-handed? 1

$$\frac{12}{120} = 0.1 = 10\%$$

b) Given that a student is right-handed, what is the fraction probability that the student was also female?

1

$$\frac{43}{98}$$

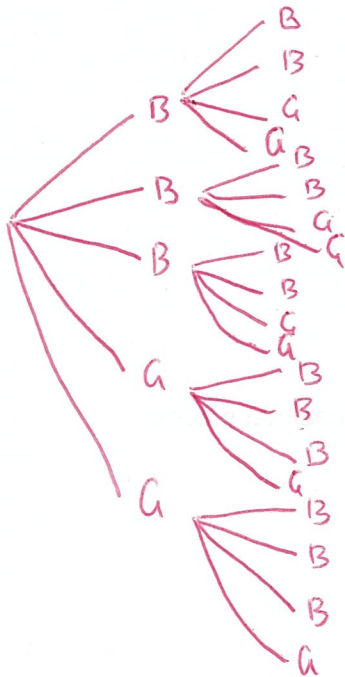
Section 2 continues on the following page

Short Answer Questions – Show working where necessary.

14. Chrissy is decorating the living room for a dinner party. In the box of decorations, there are 3 blue and 2 green items. Chrissy chooses between two decorations at random to put up.

- a) Draw a tree diagram showing the sample space for Chrissy selecting two decorations (without replacement) from the box.

2



① 1st stage/outcome correct

① 2nd stage/outcome correct

or

② All correct

- b) If the first decoration is blue, what is the fraction probability that the second decoration is green?

1

$$\frac{6}{12} = \frac{1}{2}$$

- c) What is the fraction probability of the first two decorations chosen being one of each colour (in any order)?

1

$$\frac{12}{20} = \frac{3}{5}$$

Short Answer Questions – Show working where necessary.

Surface Area And Volume

1. A cone has a volume of 1216.4 cm^3 . Find, correct to one decimal place, the radius of the cone if the height is 15 cm .

2

$$\begin{aligned}\frac{1}{3} \times \pi \times r^2 \times 15 &= 1216.4 \\ r^2 &= 77.43842911 \\ r &= \sqrt{77.43842911} \\ &= 8.8 \text{ cm}\end{aligned}$$

① Working out

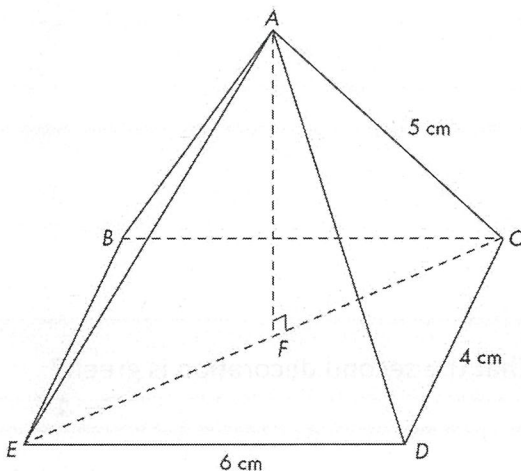
① Correct answer

or

② Correct answer

2. Calculate the volume of the rectangular pyramid, correct to two decimal places.

3



$$\begin{aligned}EC &= \sqrt{6^2 + 4^2} \\ &= \sqrt{52}\end{aligned}$$

$$FC = \frac{\sqrt{52}}{2}$$

$$\begin{aligned}AF^2 &= AC^2 - FC^2 \\ &= 5^2 - \left(\frac{\sqrt{52}}{2}\right)^2 \\ &= 12\end{aligned}$$

$$AF = \sqrt{12}$$

$$\begin{aligned}V &= \frac{1}{3} Ah \\ &= \frac{1}{3} \times (6 \times 4) \times \sqrt{12} \\ &= 27.71 \text{ cm}^3\end{aligned}$$

① Finding EC

① Finding AF

① Finding volume

or

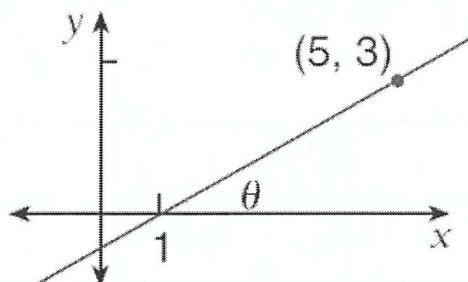
③ correct answer

Short Answer Questions – Show working where necessary.

Trigonometry

3. Find θ , the angle made between the line and the horizontal (x) axis, correct to the nearest minute.

1



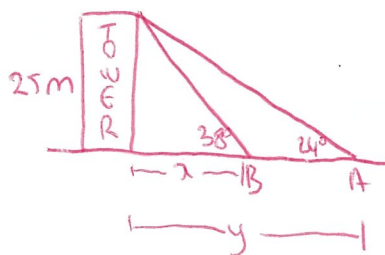
$$\tan \theta = \frac{3}{4}$$

$$\theta = \tan^{-1}\left(\frac{3}{4}\right)$$

$$= 36^{\circ}52' \quad (1)$$

4. Two points, A and B are in line with the base of a tower 25 m high. The angle of elevation of the top of the tower is 24° from A and 38° from B . How far apart are the points A and B , correct to the nearest metre.

2



$$\tan 38 = \frac{25}{x}$$

$$x = \frac{25}{\tan 38}$$

$$\tan 24 = \frac{25}{y}$$

$$y = \frac{25}{\tan 24}$$

$$\begin{aligned} y - x &= \frac{25}{\tan 24} - \frac{25}{\tan 38} \\ &= 24 \text{ m} \end{aligned}$$

① working out

① Correct answer

on

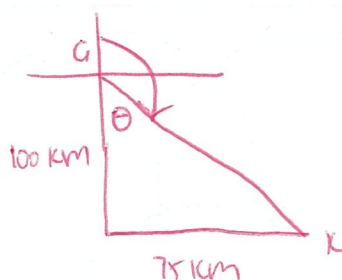
② Correct answer

5. Gru and Kevin leave home at 7 p.m. Gru drove due north at 80 km/h while Kevin drove due east at 60 km/h. Find the bearing of Kevin from Gru at 8:15 p.m.

2

$$\text{Gru: } 80 \times 1.25 = 100 \text{ km}$$

$$\text{Kevin: } 60 \times 1.25 = 75 \text{ km}$$



$$\tan \theta = \frac{75}{100}$$

$$\theta = 37^{\circ}$$

$$\begin{aligned} 180 - 37 \\ = 143^{\circ} \end{aligned}$$

① working out

① Correct answer

on

② Correct answer

Section 3 continues on the following page

Short Answer Questions – Show working where necessary.

Probability

6. One hundred citizens are surveyed regarding their use of water for their garden. 27 say that they use rain water, 42 say that they use tap water and 38 say that they do not use water at all.

a) Represent the information in the two-way table below.

① At least 4 values correct

2

② All values correct

	Rain water	No rain water	Total
Tap	7	35	42
No tap water	20	38	58
Total	27	73	100

b) What is the decimal probability that one of the citizens surveyed uses tap water or rain water? 1

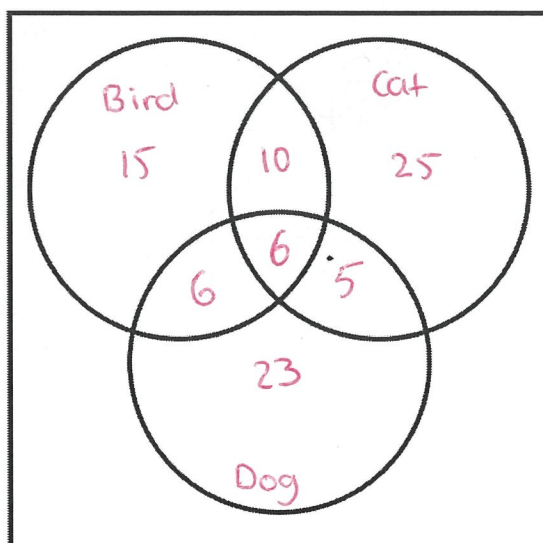
$$\frac{62}{100} = 0.62 \quad (1)$$

7. Fill in the Venn diagram to represent the information below:

2

90 students all own a dog, a cat or a bird. They were asked which pets they own.

- 37 own a bird
- 46 own a cat
- 40 own a dog
- 16 own both a cat and a bird
- 11 own both a dog and a cat
- 12 own both a bird and a dog
- 6 own a dog, a cat and a bird



① At least 4 values correct

② All values correct

End of Exam :)